

MATH 2630-090: Final Exam – Summer 2024

July 31, 2024

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Name (*please print*): _____

Instructions: Please read the following instructions carefully.

- **Read all of the directions** for each problem.
- You have **120 minutes** to finish the exam.
- **Show work to receive credit.** Answers that are wrong, but still have work might receive partial credit. If you provide the answer to a computation without work, I will not give partial credit.
- Provide **exact** solutions.
- Do **not** write in the table below.

Problem	Points	Score
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total	200	

1. (20 points) Find $\frac{\partial f}{\partial t}$ where $f(x, y) = e^{xy} + x^2y$, $x = 2u - v$, $y = u^2 + v^2$, $u = 3s - 4t$, $v = \sqrt{s}$.

2. (20 points) Evaluate the following limit. Write “Does not exist” if applicable.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2 + (x - y)^2}.$$

3. (20 points) Consider the following surface.

$$f(x, y) = 4xy - x^4 - y^4.$$

Find *and* classify **ALL** relative maximum, minimums, and saddle points.

4. (20 point) Consider the following surface.

$$z = y^2 \ln x$$

- (i) Find the slope of the tangent line to the surface at $(e, 1)$ in the direction of $\vec{u} = \langle 2, 3 \rangle$.
- (ii) Find the equation of the tangent plane to the surface at the point $(e, 1, 1)$.

5. (20 points) Evaluate the following double integral.

$$\int_0^2 \int_{y/2}^1 \cos(x^2) \, dx \, dy.$$

6. (20 points) Set up but **DO NOT EVALUATE** an expression for the volume of the solid bounded by the sphere with radius 5, centered at the origin and the below the xy -plane.

7. (20 points) Set up **BUT DO NOT EVALUATE** the following double integral using the transformation $u = \frac{y}{x}$ and $v = xy$.

$$\iint_R e^{xy} \, dA$$

Where R is bounded by the lines $y = \frac{1}{2}x$, $y = x$, and the hyperbolas $y = \frac{1}{x}$, and $y = \frac{2}{x}$.

8. (20 points) Evaluate the following line integral where C is the curve formed by the circle $x^2 + y^2 = 1$ and $y = |x|$.

$$\oint_C \sqrt{x} \, dx + (xy + e^{\tan y}) \, dy$$

9. (20 points) Consider the vector field $F(x, y) = (2xy^2 + 2y)\hat{i} + (2x^2y + 2x)\hat{j}$

(i) Find the divergence of F at the point $(2, 3)$.

(ii) Find the curl of F at the point $(2, 3)$

(iii) Find the amount of work to go through F from $(1, 1)$ to $(2, 2)$.

10. (20 points) Evaluate the line integral where C is line segment from $(1, 2, 3)$ to $(2, 5, 4)$.

$$\int_C x + y + z \, ds$$